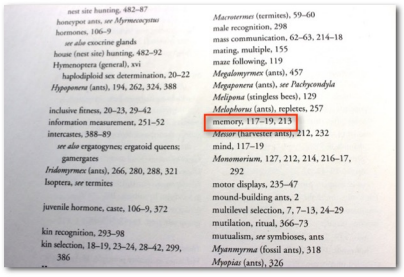
Pre-processing T

- If the Text will be searched many times, then there is very little cost of pre-processing
- An algorithm that pre-processes T is an *offline* algorithm, while one that does NOT pre-process it is an *online* algorithm.

149

Indexing



Key terms ordered alphabetically, with associated page #s

Grocery store items grouped into aisles

150

Indexing DNA

Index of T

T: C G T G C G T G C T T

151

Indexing DNA

Index of T

C G T G C : 0

T: C G T G C G T G C T T

152

Indexing DNA

Index of T

C G T G C : **0**
G T G C G : **1**

T: C G T G C G T G C T T

153

Indexing DNA

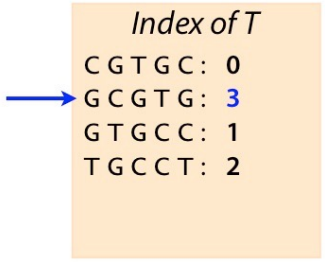
Index of T

C G T G C : **0**
G T G C G : **1**
T G C G T : **2**

T: C G T G C G T G C T T

154

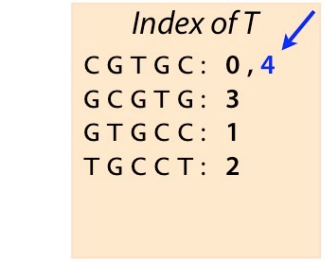
Indexing DNA



T: C G T G C G T G C T T

155

Indexing DNA



T: C G T G C G T G C T T

156

Indexing DNA

Index of T

CGTGC: 0,4
GCGTG: 3
GTGCC: 1
GTGCT: 5
TGCCT: 2

T: C G T G C G T G C T T

157

Indexing DNA

Index of T

CGTGC: 0,4
GCGTG: 3
GTGCC: 1
GTGCT: 5
TGCCT: 2
TGCTT: 6

T: C G T G C G T G C T T

158

Indexing DNA

k-mer: substring of length k

Index of T	
C G T G C :	**0 , 4**
G C G T G :	**3**
G T G C C :	**1**
G T G C T :	**5**
T G C C T :	**2**
T G C T T :	**6**

5-mer index

T: C G T G C G T G C T T

159

Querying the index

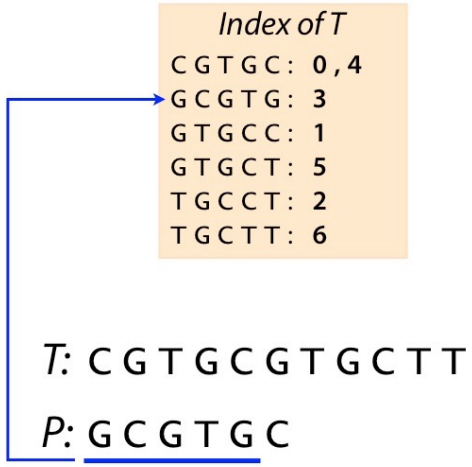
Index of T	
C G T G C :	**0 , 4**
G C G T G :	**3**
G T G C C :	**1**
G T G C T :	**5**
T G C C T :	**2**
T G C T T :	**6**

T: C G T G C G T G C T T

P: G C G T G C

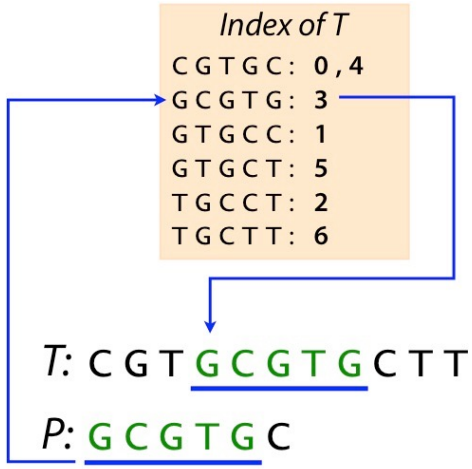
160

Querying the index



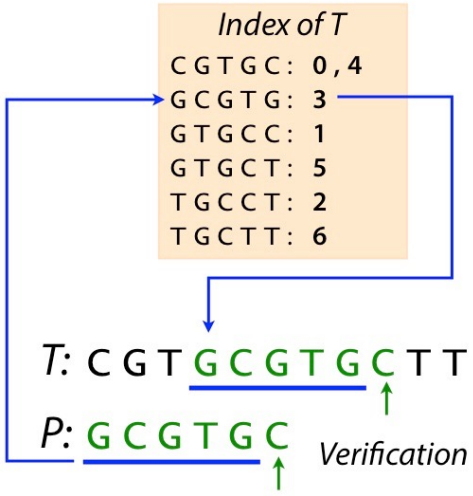
161

Querying the index



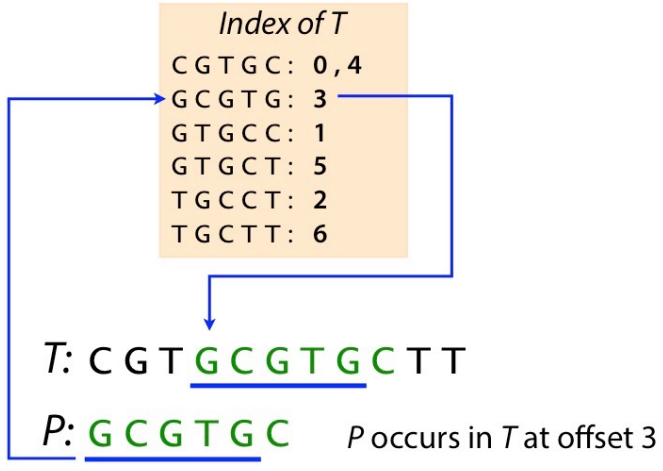
162

Querying the index



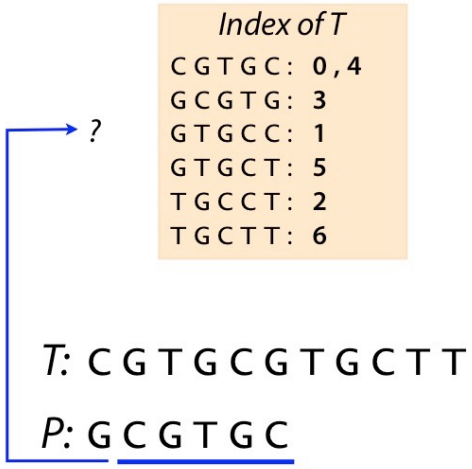
163

Querying the index



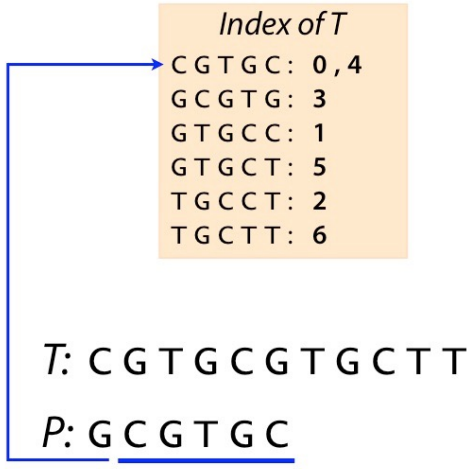
164

Querying the index



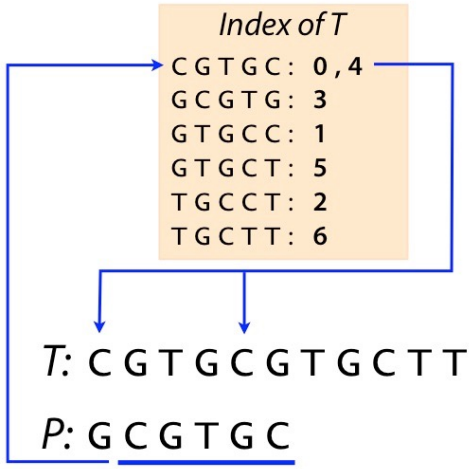
165

Querying the index



166

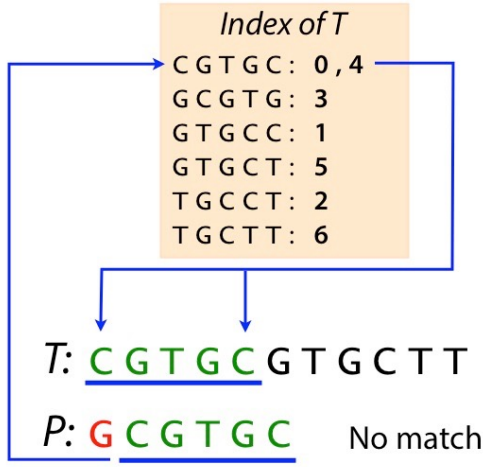
Querying the index



Two hits

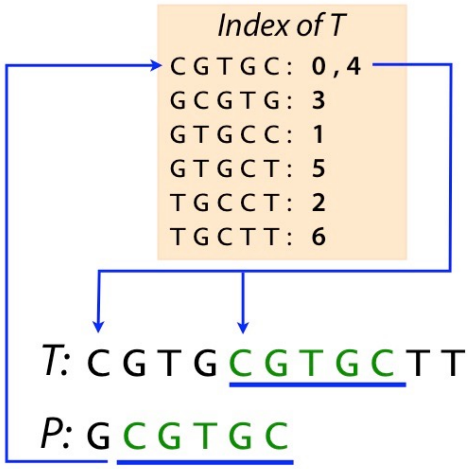
167

Querying the index



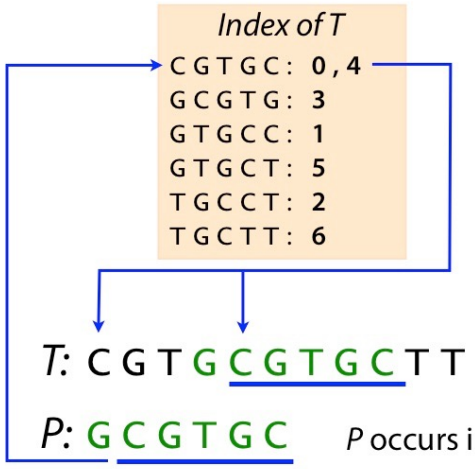
168

Querying the index



169

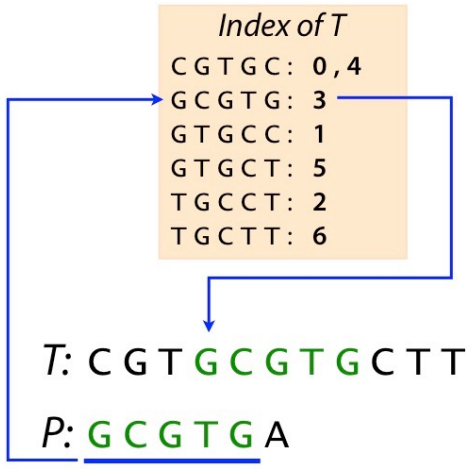
Querying the index



P occurs in *T* at offset 3

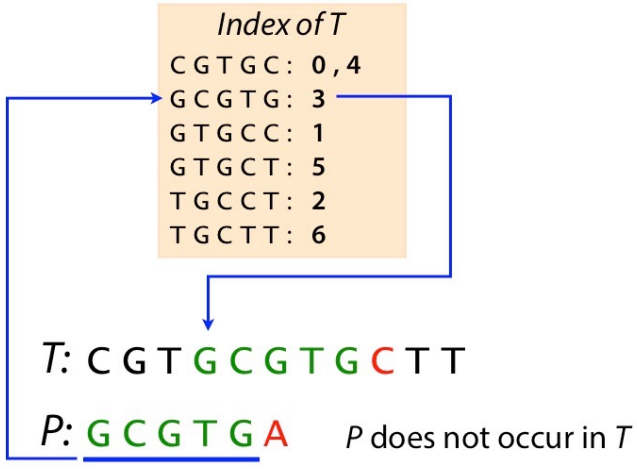
170

Querying the index



171

Querying the index



172

Querying the index

Index of T

CGTGC: 0, 4
GCGTG: 3
GTGCC: 1
GTGCT: 5
TGCCT: 2
TGCTT: 6

T: C G T G C G T G C T T

P: G C G T A C

173

Querying the index

Index of T

CGTGC: 0, 4
GCGTG: 3
GTGCC: 1
GTGCT: 5
TGCCT: 2
TGCTT: 6

T: C G T G C G T G C T T

P: G C G T A C

P does not occur in *T*

174

Data structures

Index of T

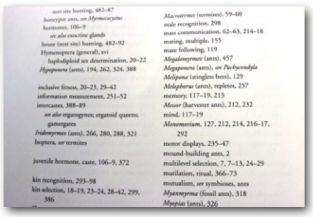
CGTGC: **0,4**
GCGTG: **3**
GTGCC: **1**
GTGCT: **5**
TGCCT: **2**
TGCTT: **6**

Multimap

T: C G T G C G T G C T T

175

Data structures



Multimap

T: C G T G C G T G C T T

176

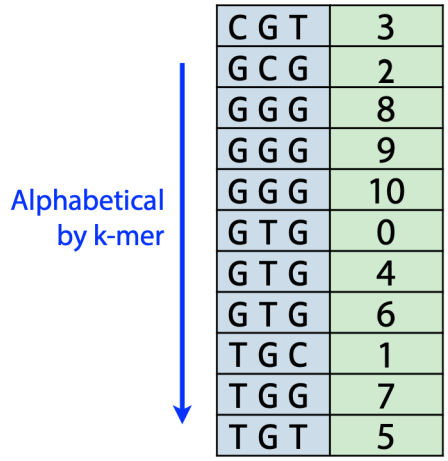
Data structures

G T G	0
T G C	1
G C G	2
C G T	3
G T G	4
T G T	5
G T G	6
T G G	7
G G G	8
G G G	9
G G G	10

T: G T G C G T G T G G G G G

177

Data structures

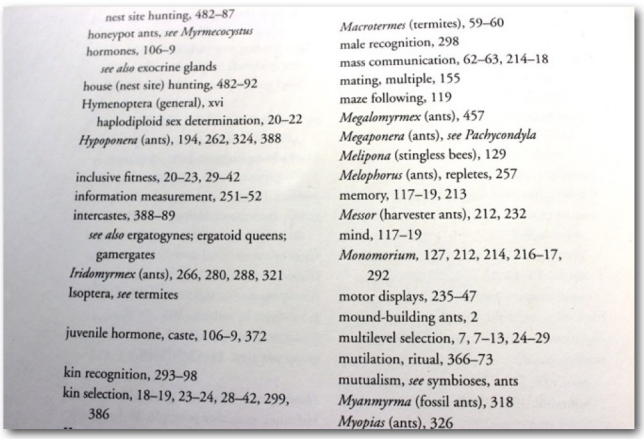


C G T	3
G C G	2
G G G	8
G G G	9
G G G	10
G T G	0
G T G	4
G T G	6
T G C	1
T G G	7
T G T	5

T: G T G C G T G T G G G G G

178

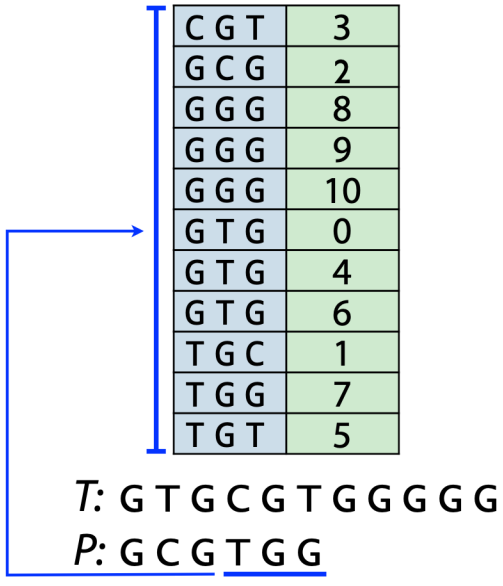
Data structures



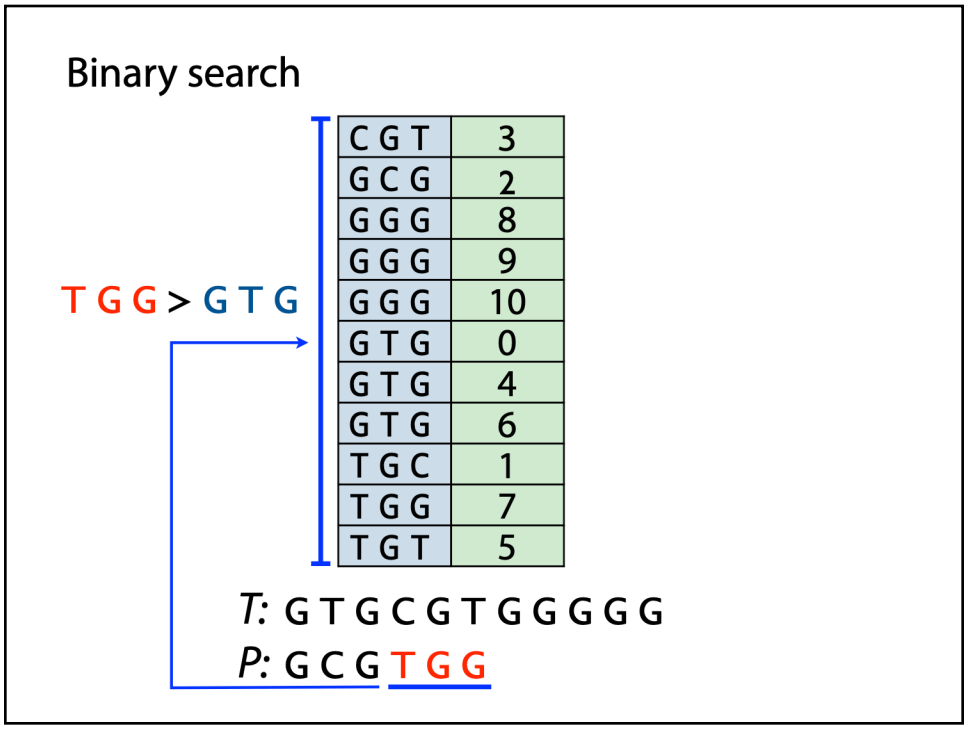
Binary Search

179

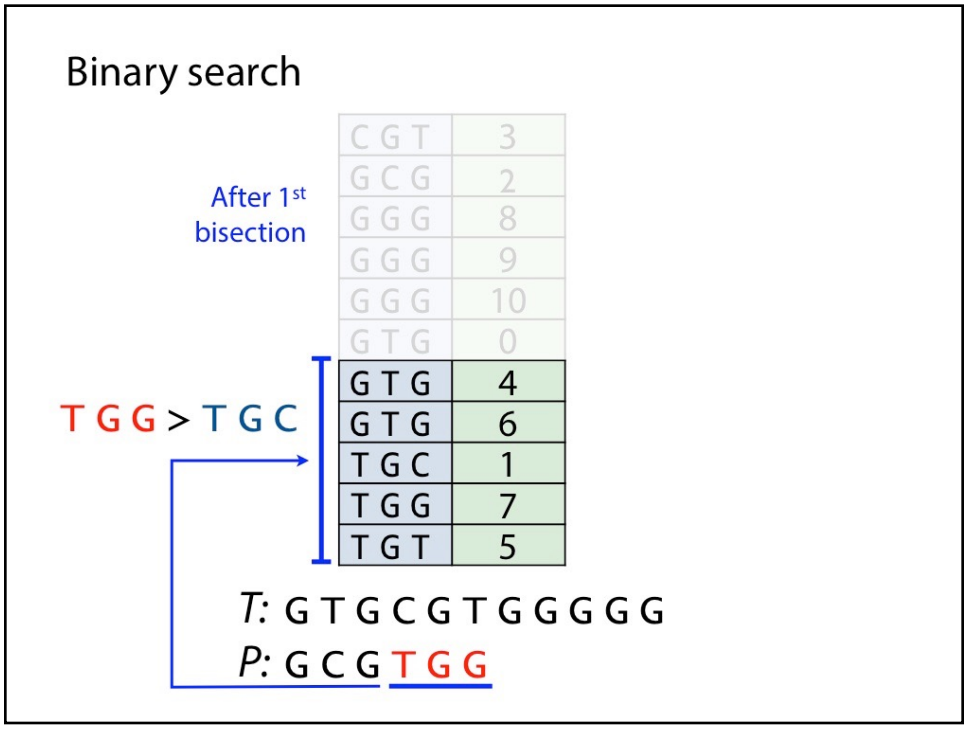
Binary search



180

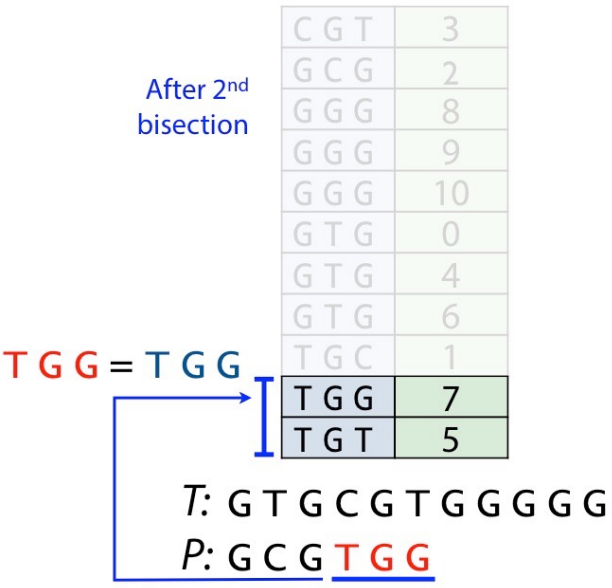

Binary search
C G T 3
G C G 2
G G G 8
G G G 9
G G G 10
G T G 0
G T G 4
G T G 6
T G C 1
T G G 7
T G T 5
T G G > G T G
T: G T G C G T G G G G G
P: G C G T G G

181


Binary search
After 1st bisection
C G T 3
G C G 2
G G G 8
G G G 9
G G G 10
G T G 0
G T G 4
G T G 6
T G C 1
T G G 7
T G T 5
T G G > T G C
T: G T G C G T G G G G G
P: G C G T G G

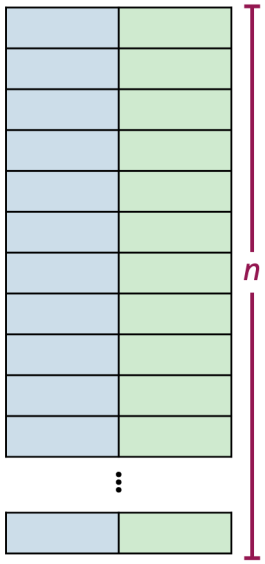
182

Binary search



183

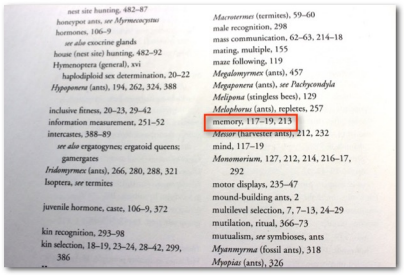
Binary search



~ $\log_2(n)$ bisections per query

184

Indexing



nest site hunting, 482–87
honeypot ants, *see Myrmecocystus*
hormones, 106–9
see also exocrine glands
house (nest site) hunting, 482–92
Hymenoptera (general), xvi
haplodiploid sex determination, 20–22
Hypoponera (ants), 194, 262, 324, 388

inclusive fitness, 20–23, 29–42
information measurement, 251–52
intercastes, 388–89
see also ergatogynes; ergatoid queens; gamergates
Iridomyrmex (ants), 266, 280, 288, 321
Isoptera, *see* termites

juvenile hormone, caste, 106–9, 372

kin recognition, 293–98
kin selection, 18–19, 23–24, 28–42, 299, 386

Macrotermes (termites), 59–60
male recognition, 298
mass communication, 62–63, 214–18
mating, multiple, 155
maze following, 119
Megalomyrmex (ants), 457
Megaponera (ants), *see Pachycondyla*
Melipona (stingless bees), 129
Melophorus (ants), repletes, 257
memory, 117–19, 213
Messor (harvester ants), 212, 232
mind, 117–19
Monomorium, 127, 212, 214, 216–17, 292
motor displays, 235–47
mound-building ants, 2
multilevel selection, 7, 7–13, 24–29
mutilation, ritual, 366–73
mutualism, *see* symbioses, ants
Myanmyrma (fossil ants), 318
Myopias (ants), 326

Key terms ordered alphabetically, with associated page #s

Grocery store items grouped into aisles

185

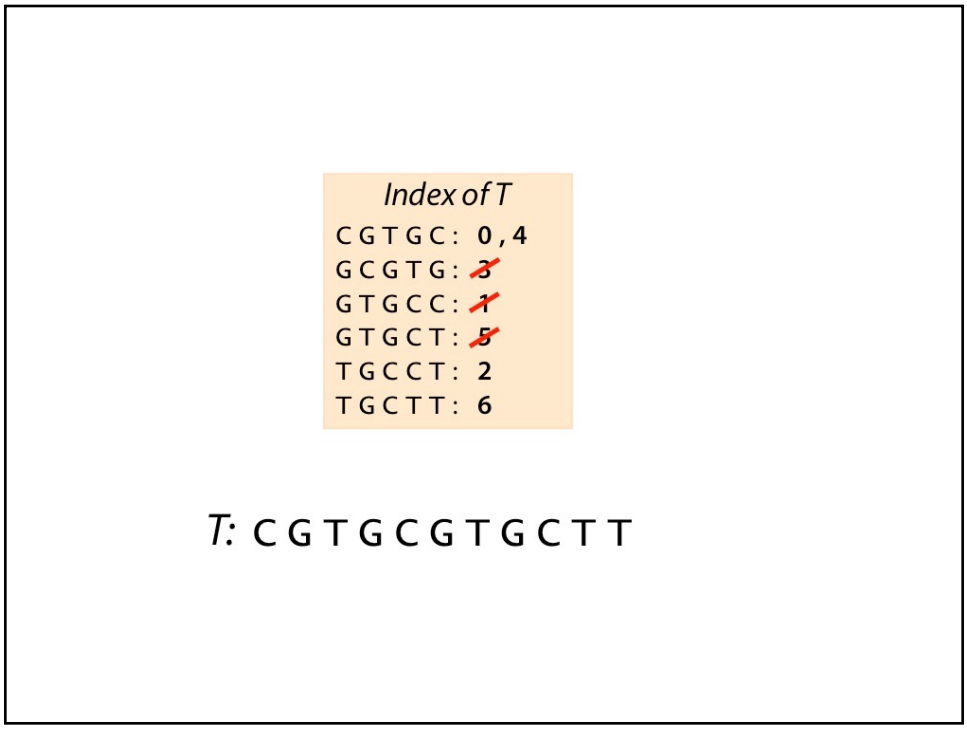
Kmer Index Variations

Index of T

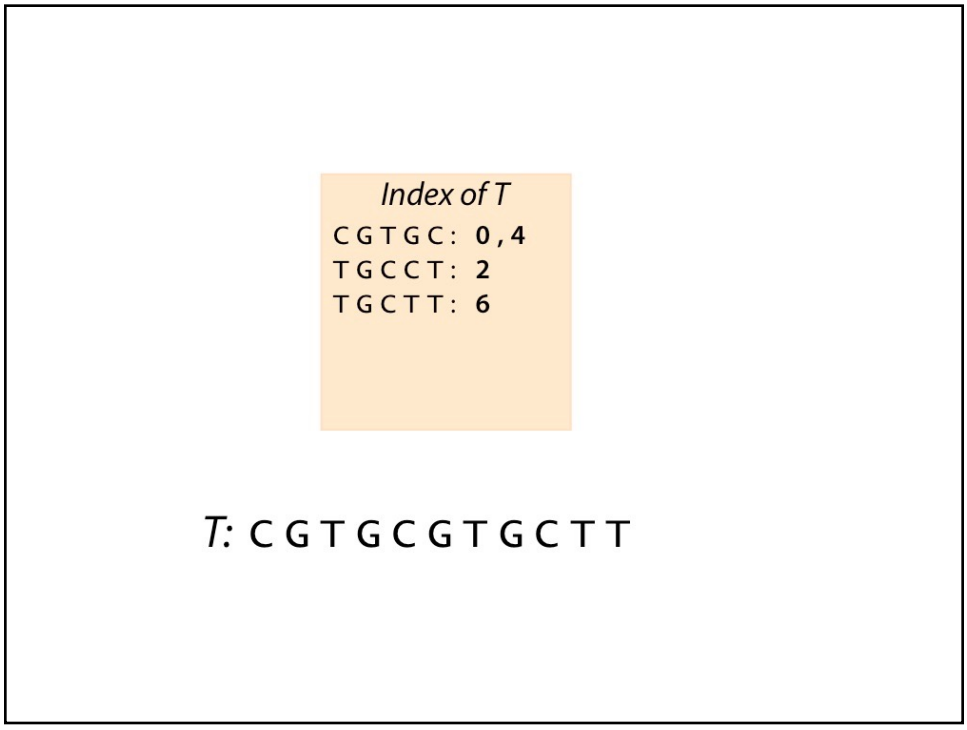
CGTGC: **0,4**
GCGTG: **3**
GTGCC: **1**
GTGCT: **5**
TGCCT: **2**
TGCTT: **6**

T: C G T G C G T G C T T

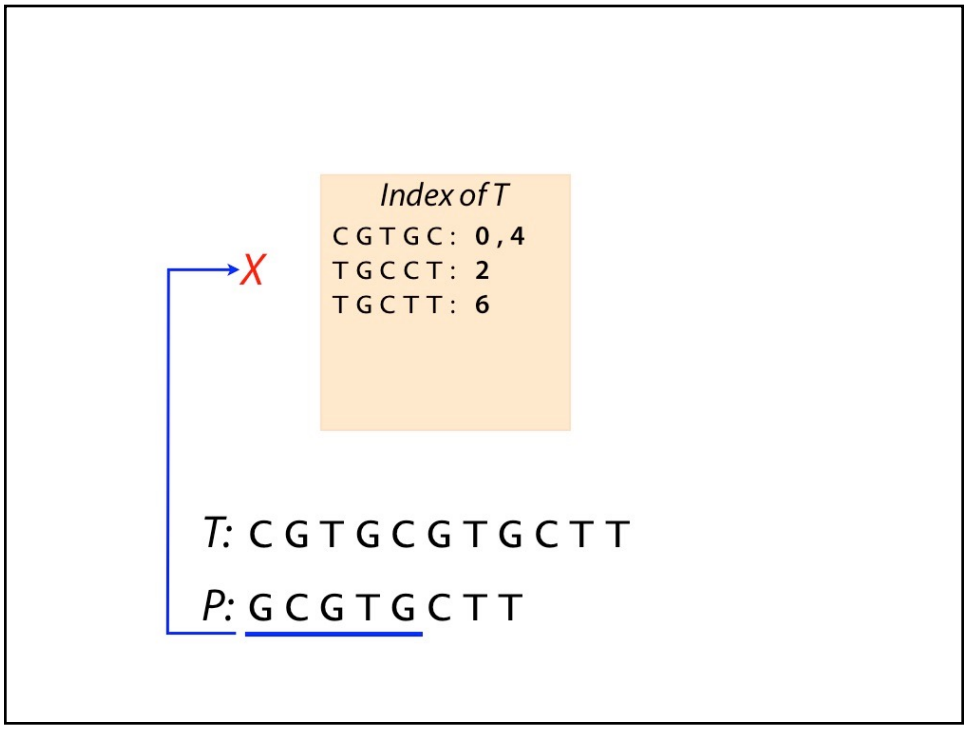
186


Index of T
CGTGC: 0,4
GCGTG: 3
GTGCC: 1
GTGCT: 5
TGCCT: 2
TGCTT: 6
T: C G T G C G T G C T T

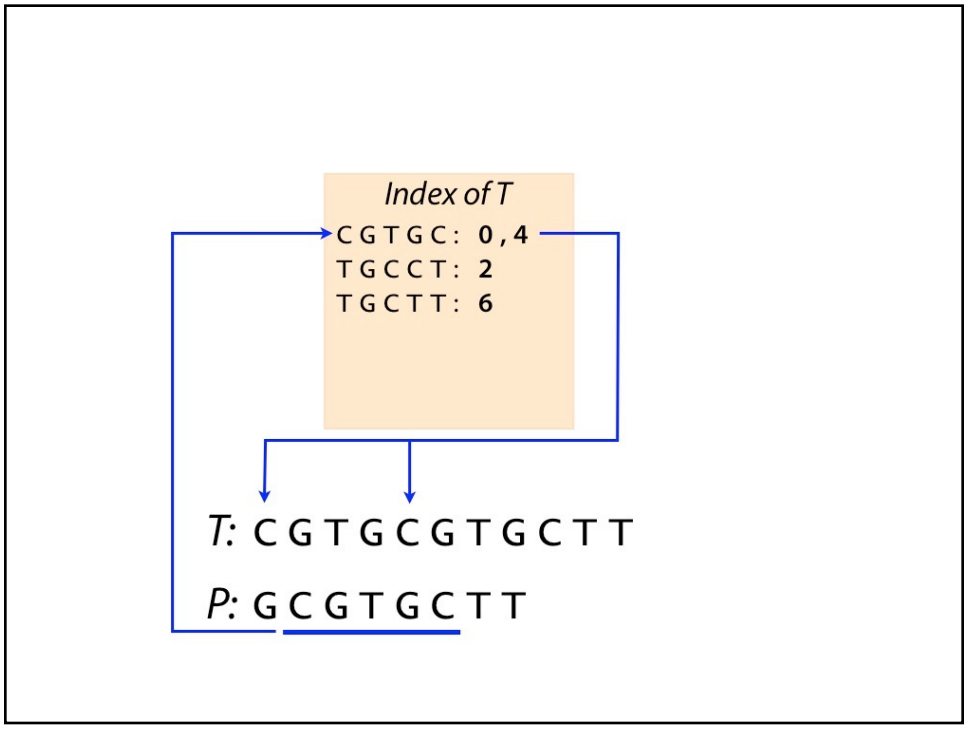
187


Index of T
CGTGC: 0,4
TGCCT: 2
TGCTT: 6
T: C G T G C G T G C T T

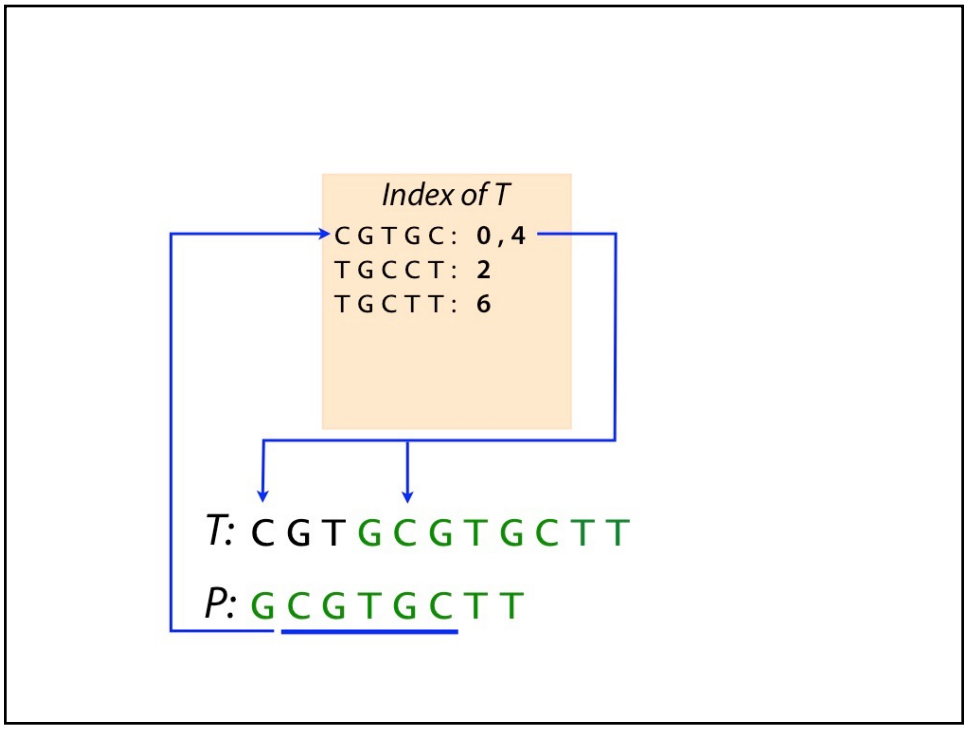
188


Index of T
CGTGC: 0,4
TGCCT: 2
TGCTT: 6
X
T: C G T G C G T G C T T
P: G C G T G C T T

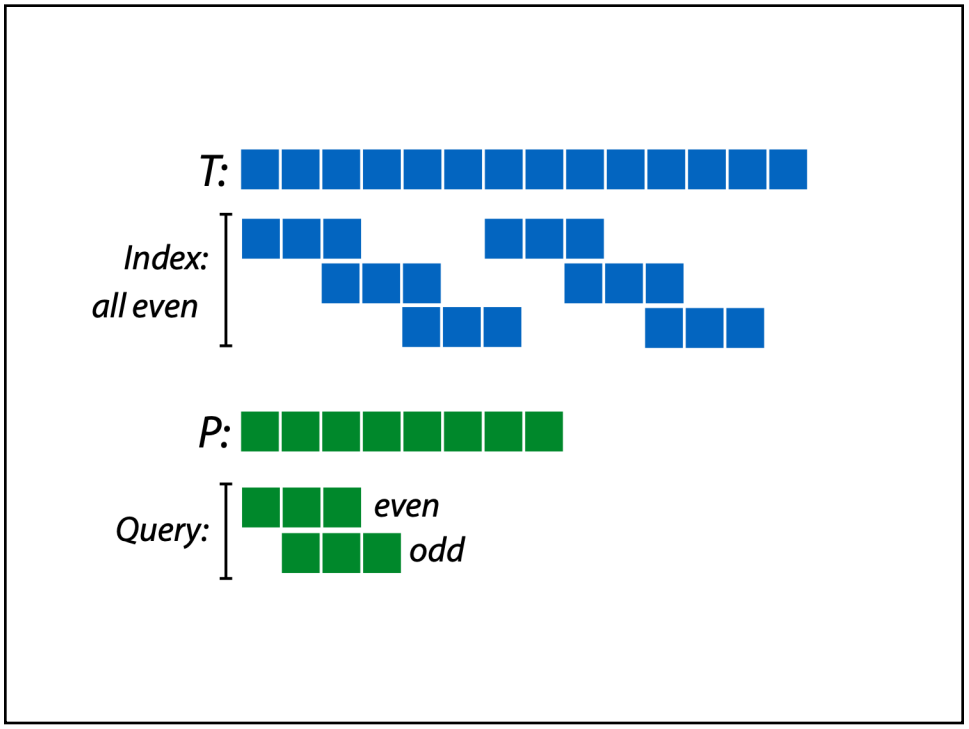
189


Index of T
CGTGC: 0,4
TGCCT: 2
TGCTT: 6
T: C G T G C G T G C T T
P: G C G T G C T T

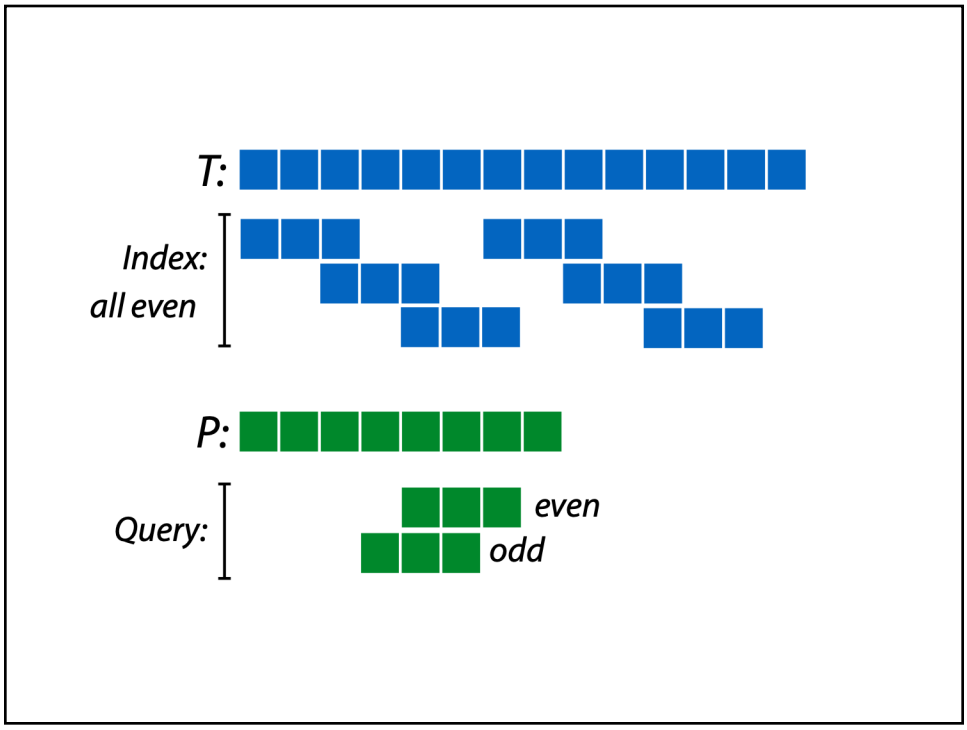
190


Index of T
CGTGC: 0,4
TGCCT: 2
TGCTT: 6
T: C G T G C G T G C T T
P: G C G T G C T T

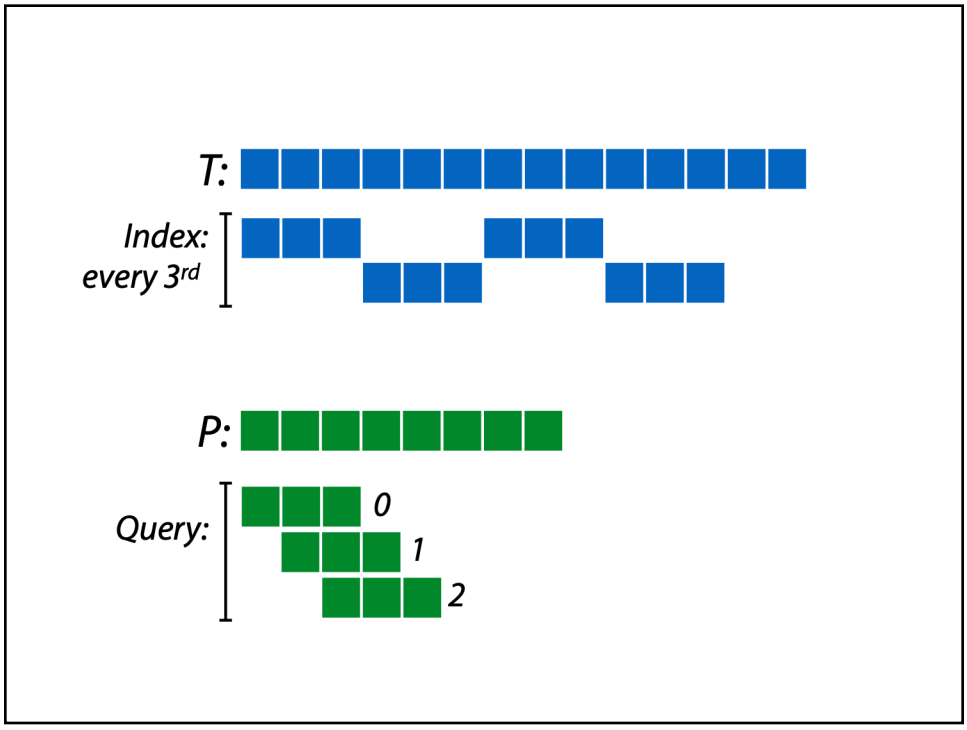
191


T:
Index:
all even
P:
Query:
even
odd

192



193



194

Kmer Index Variations -- Subsequence

Index of T

C G G G T : **0**

T: C G T G C G T G C T T

195

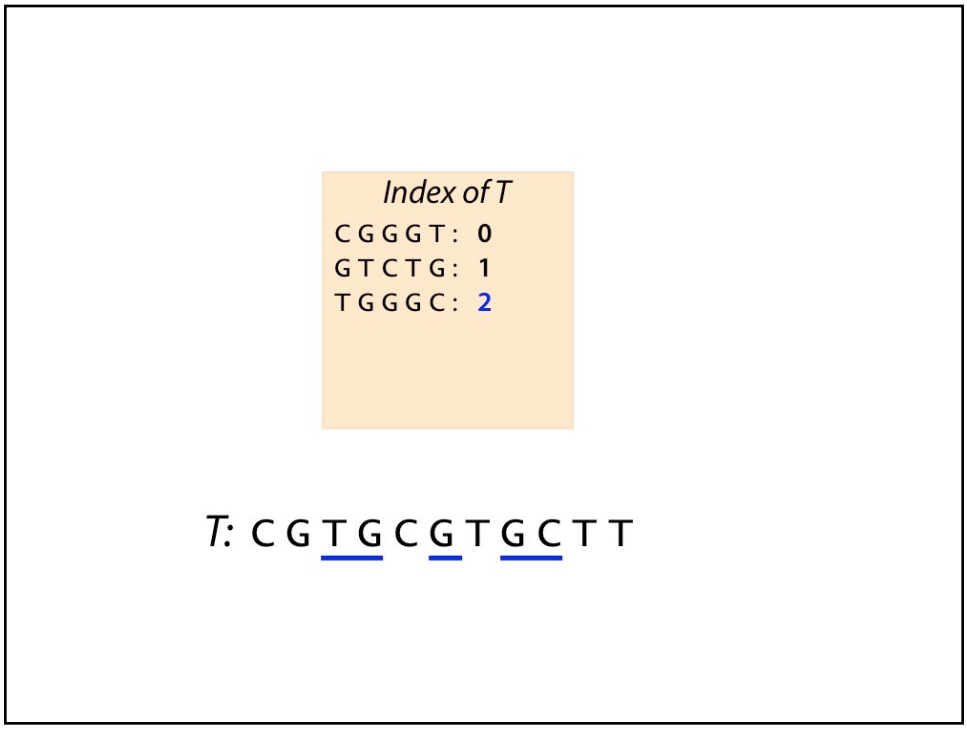
Index of T

C G G G T : **0**

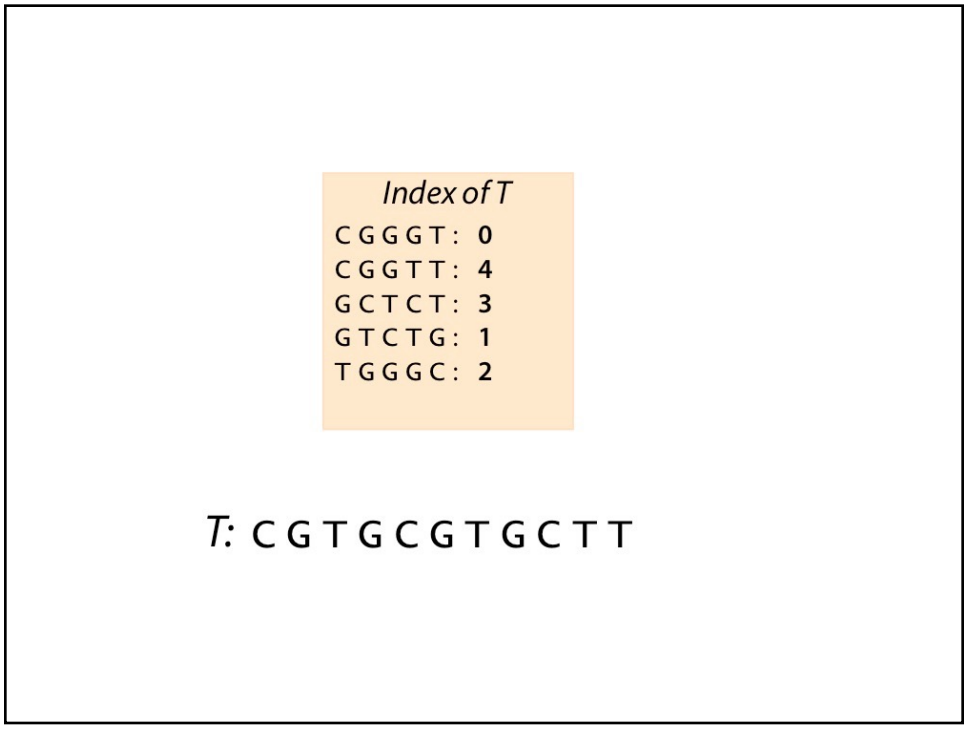
G T C T G : **1**

T: C G T G C G T G C T T

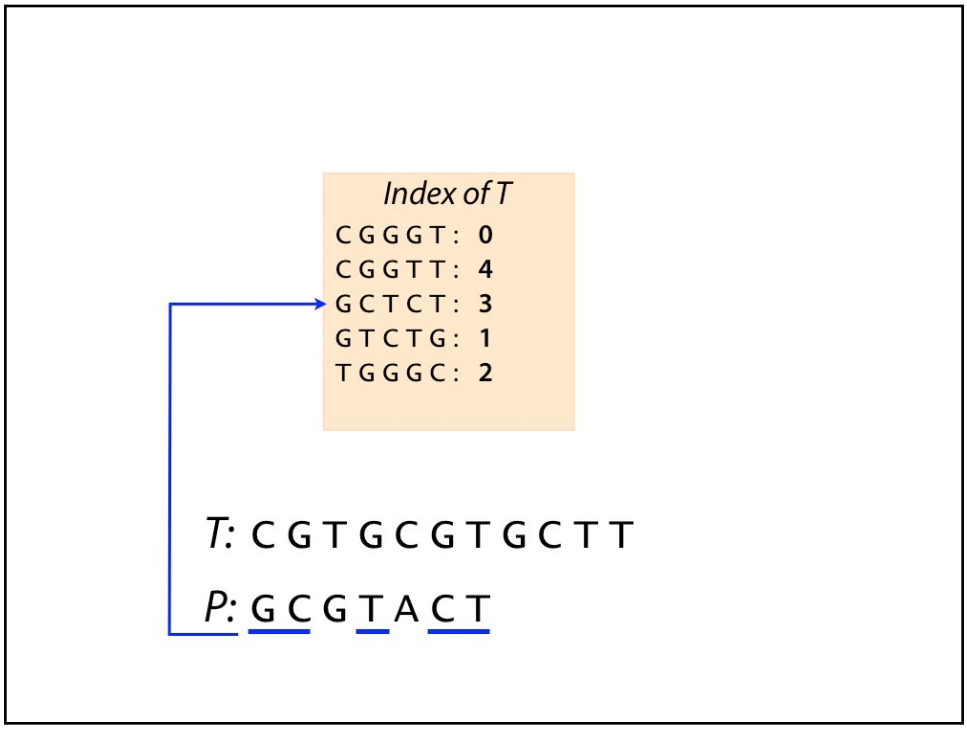
196


Index of T
C G G G T : 0
G T C T G : 1
T G G G C : 2
T: C G T G C G T G C T T

197


Index of T
C G G G T : 0
C G G T T : 4
G C T C T : 3
G T C T G : 1
T G G G C : 2
T: C G T G C G T G C T T

198



199

Suffix index

T: GTTATAGCTGATCGCGGCGTAGCGG
GTTATAGCTGATCGCGGCGTAGCGG
TTATAGCTGATCGCGGCGTAGCGG
TATAGCTGATCGCGGCGTAGCGG
ATAGCTGATCGCGGCGTAGCGG
TAGCTGATCGCGGCGTAGCGG
AGCTGATCGCGGCGTAGCGG
GCTGATCGCGGCGTAGCGG
CTGATCGCGGCGTAGCGG
TGATCGCGGCGTAGCGG
GATCGCGGCGTAGCGG
ATCGCGGCGTAGCGG
TCGCGGCGTAGCGG
CGCGGCGTAGCGG
GCGGCGTAGCGG
CGGCGTAGCGG
GGCGTAGCGG
GCGTAGCGG
CGTAGCGG
GTAGCGG
TAGCGG
AGCGG
GCGG
CGG
GG
G

200

Suffix index

T = abaaba

abaaba
baaba
aaba
aba
ba
a

Alphabetical order →

a
aaba
aba
abaaba
ba
baaba

201

Suffix index

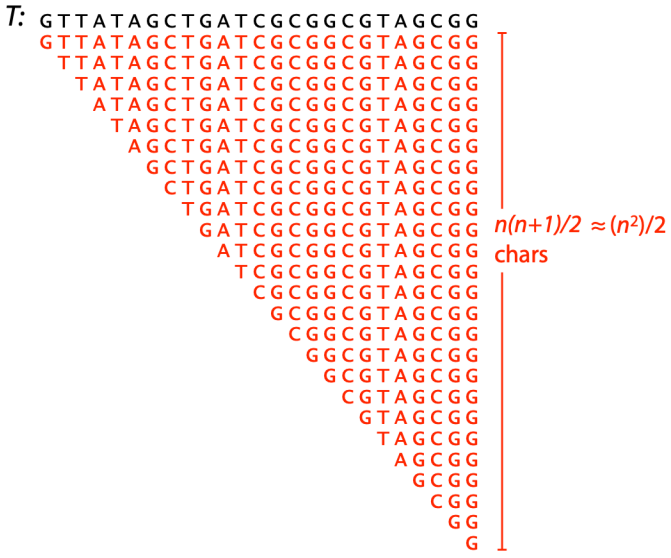
Querying uses binary search

P = ab

a
aaba
aba
abaaba
ba
baaba

202

Suffix index



203

Suffix index

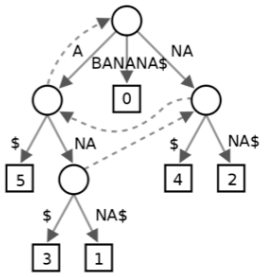
T = abaaba

Suffix array is m integers long

5	a
2	aaba
3	aba
0	abaaba
4	ba
1	baaba

SuffixArray(T)

204



6	$
5	A$
3	ANA$
1	ANANA$
0	BANANA$
4	NA$
2	NANA$

$ B A N A N A
A $ B A N A N
A N A $ B A N
A N A N A $ B
B A N A N A $
N A $ B A N A
N A N A $ B A

Suffix tree
≥ 45 GB

Suffix array
≥ 12 GB

FM Index
~ 1 GB

205

BOW TIE

Software

Open Access

Ultrafast and memory-efficient alignment of short DNA sequences to the human genome

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Accepted: 4 March 2009

Abstract

Bowtie is an ultrafast, memory-efficient alignment program for aligning short DNA sequence reads to large genomes. For the human genome, Burrows-Wheeler indexing allows Bowtie to align more than 25 million reads per CPU hour with a memory footprint of approximately 1.3 gigabytes. Bowtie extends previous Burrows-Wheeler techniques with a novel quality-aware backtracking algorithm that permits mismatches. Multiple processor cores can be used simultaneously to achieve even greater alignment speeds. Bowtie is open source http://bowtie.cbcb.umd.edu.

Fast gapped-read alignment with Bowtie 2

Ben Langmead[1,2] & Steven L Salzberg[1–3]

As the rate of sequencing increases, greater throughput is demanded from read aligners. The full-text minute index is often used to make alignment very fast and memory-efficient, but the approach is ill-suited to finding longer, gapped alignments. Bowtie 2 combines the strengths of the full-text minute index with the flexibility and speed of hardware-accelerated dynamic programming algorithms to achieve a combination of high speed, sensitivity and accuracy.

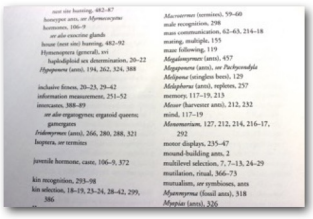
206

Why is it useful?

- Check the accuracy and completeness of *de novo* assemblies
- Map RNASeq data to genomes
- Map metagenomic reads to reference genomes
- Identify insertions

207

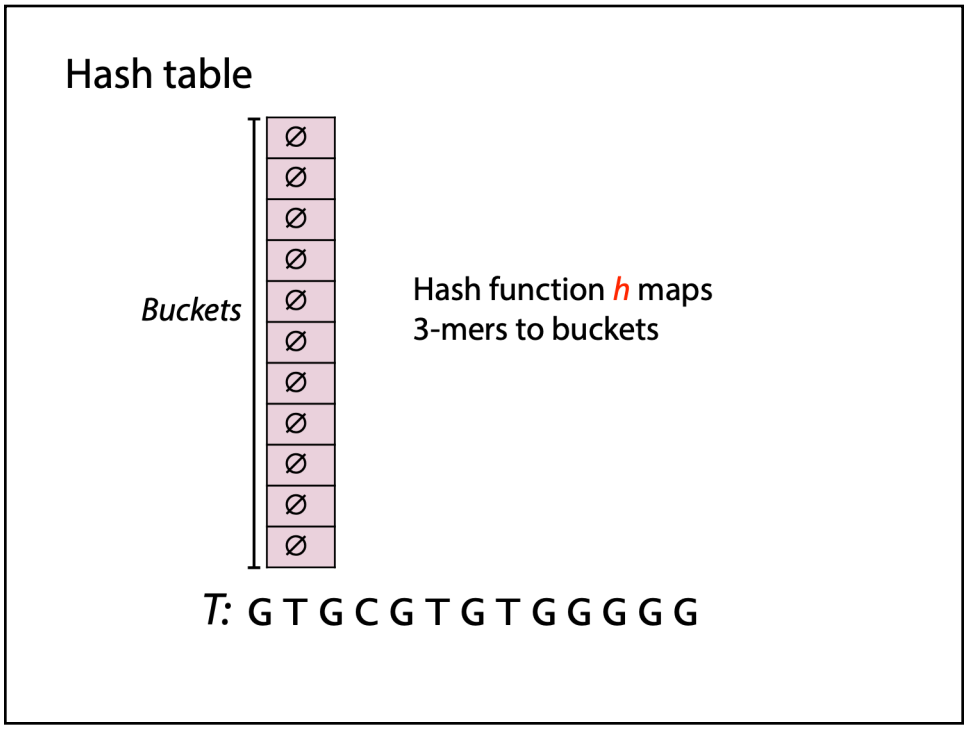
Data structures



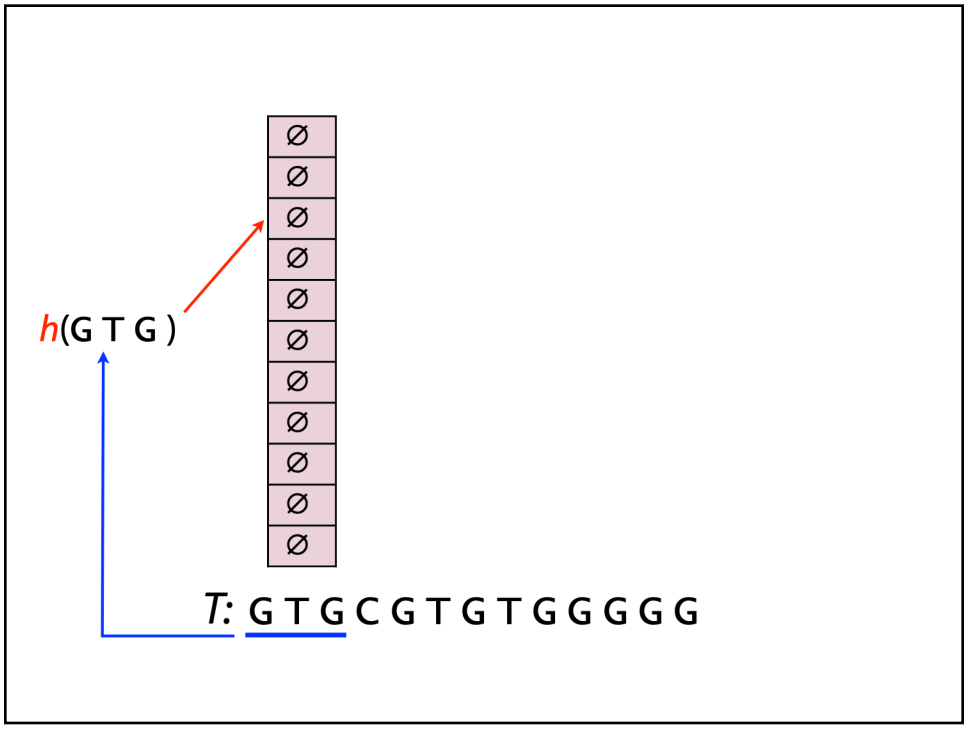
Multimap

T: C G T G C G T G C T T

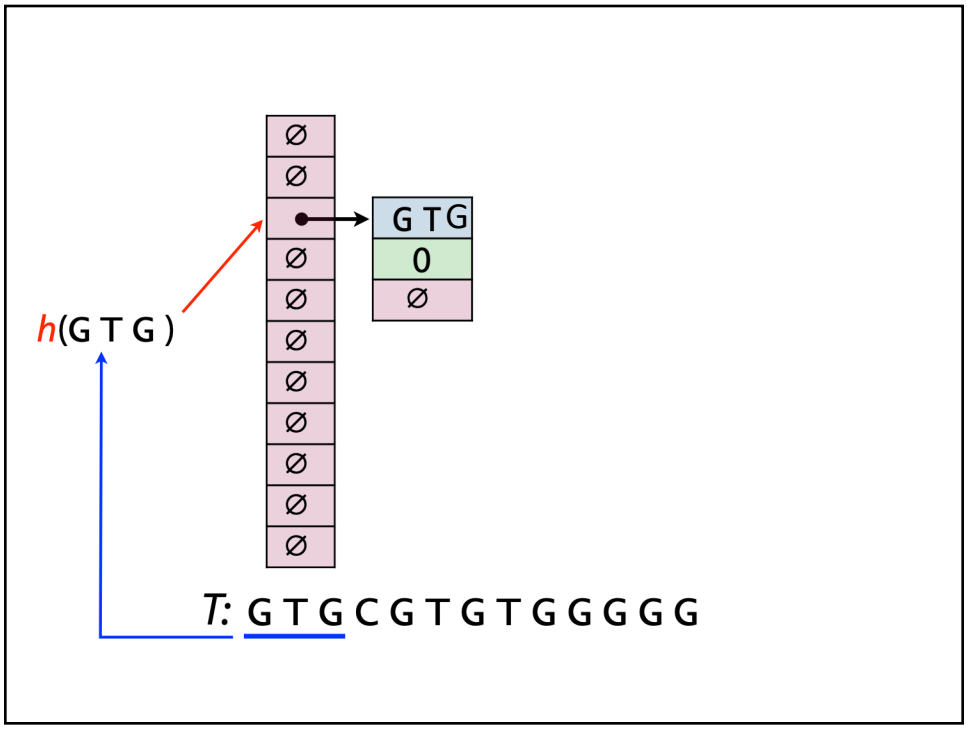
208


Hash table
Buckets
Hash function h maps
3-mers to buckets
T: G T G C G T G T G G G G G

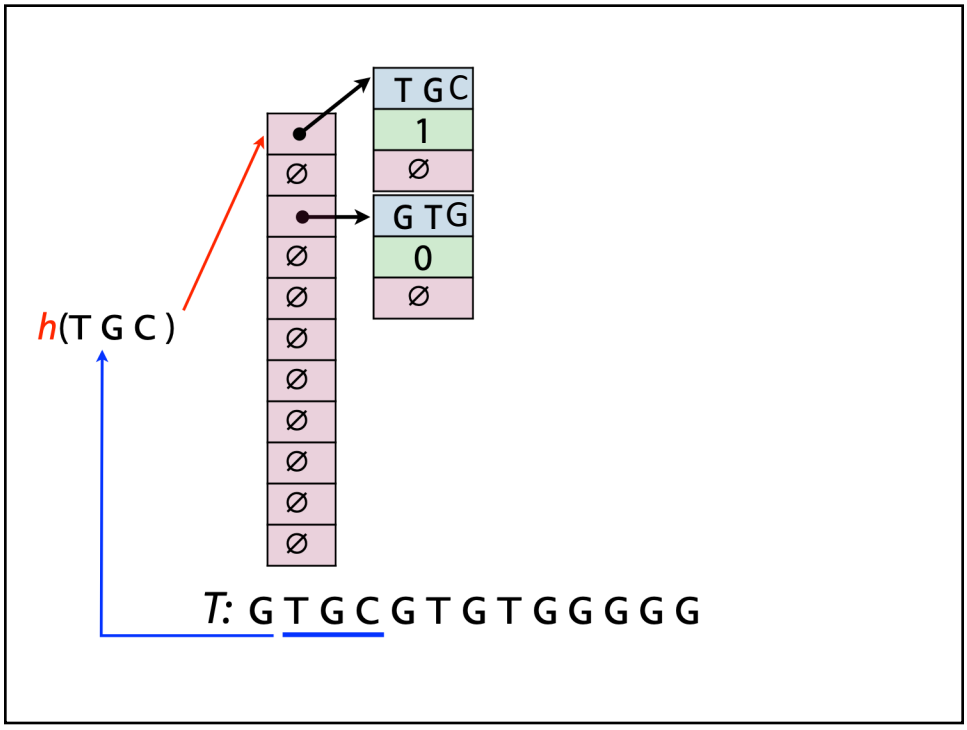
209


h(G T G)
T: G T G C G T G T G G G G G

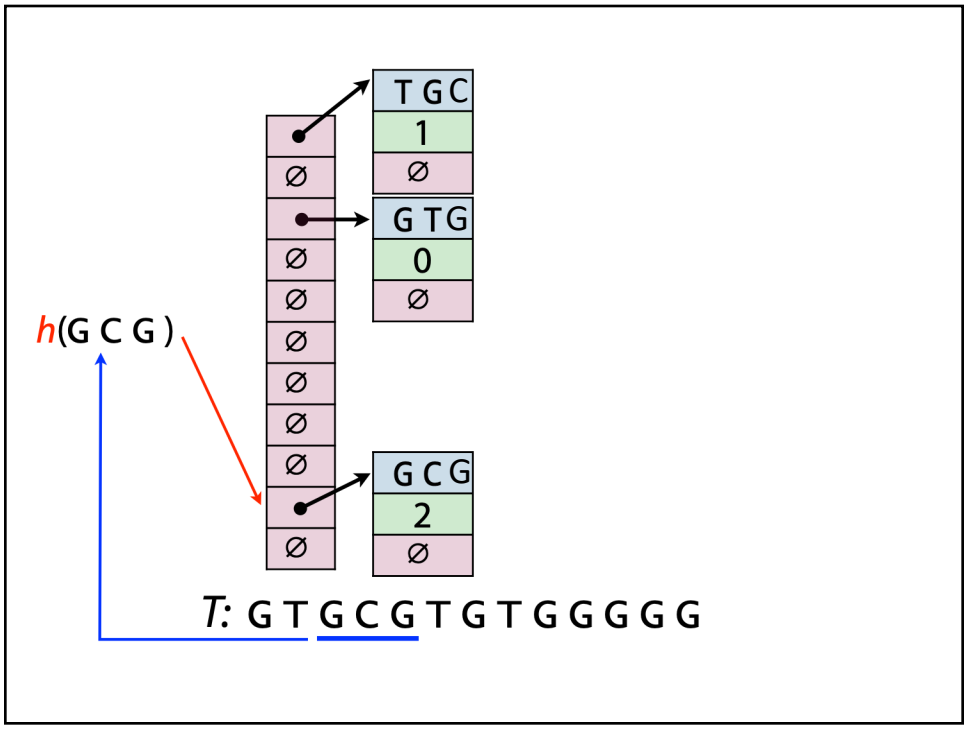
210


Ø
Ø
G T G
0
Ø
Ø
Ø
Ø
Ø
Ø
Ø
Ø
Ø
Ø
h(G T G)
T: G T G C G T G T G G G G

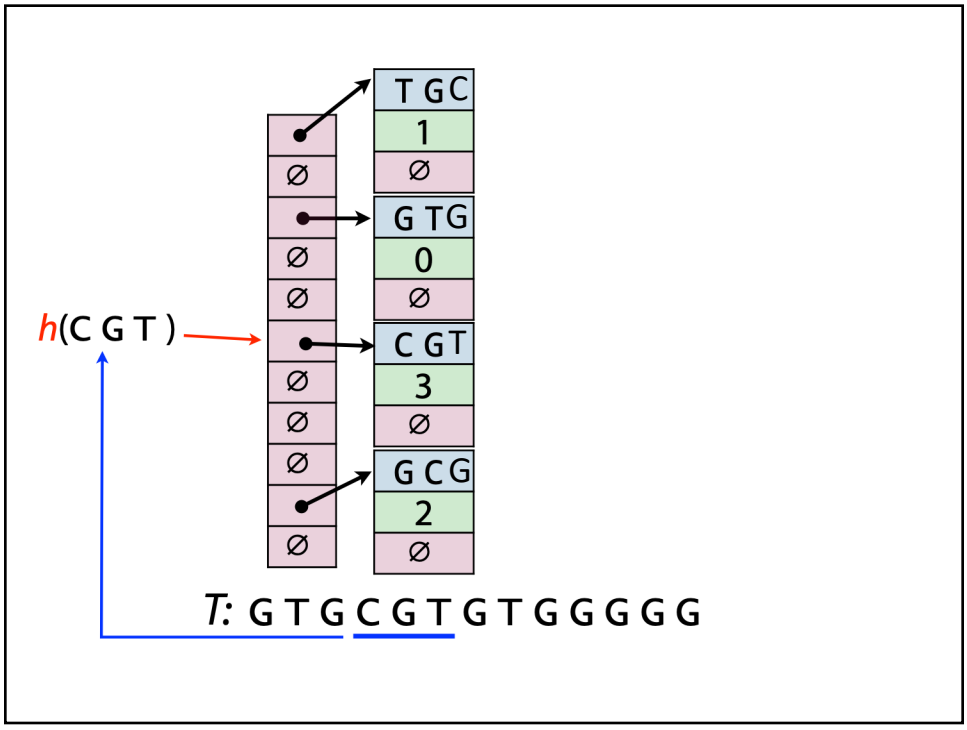
211


T G C
1
Ø
G T G
0
Ø
Ø
Ø
Ø
Ø
Ø
Ø
Ø
Ø
Ø
h(T G C)
T: G T G C G T G T G G G G

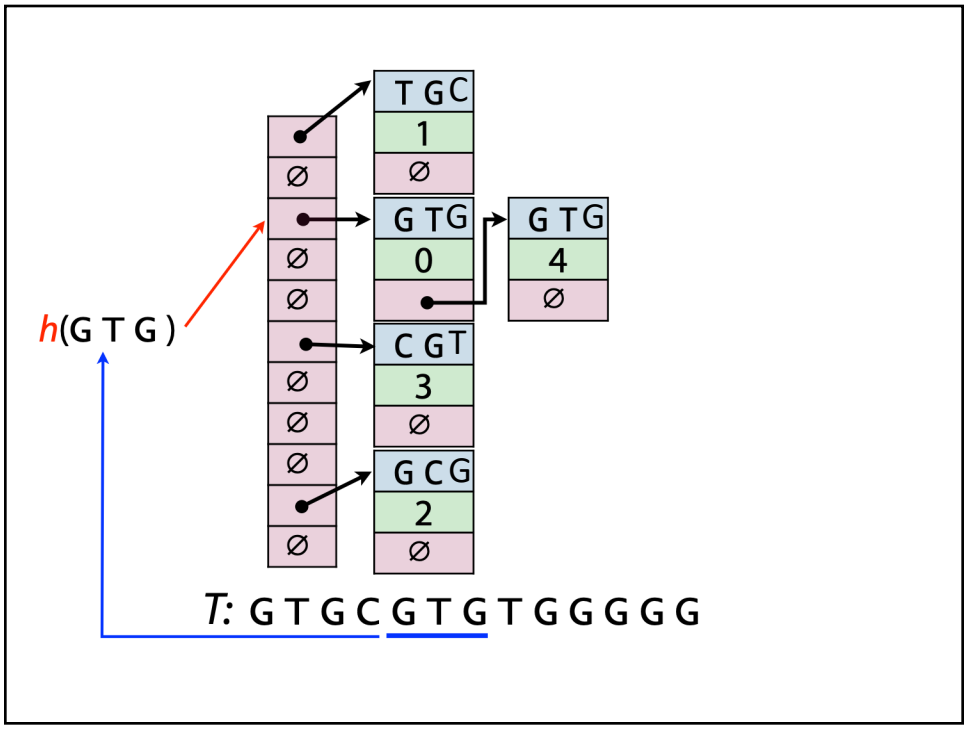
212


T G C
1
∅
G T G
0
∅
G C G
2
∅
h(G C G)
T: G T G C G T G T G G G G G

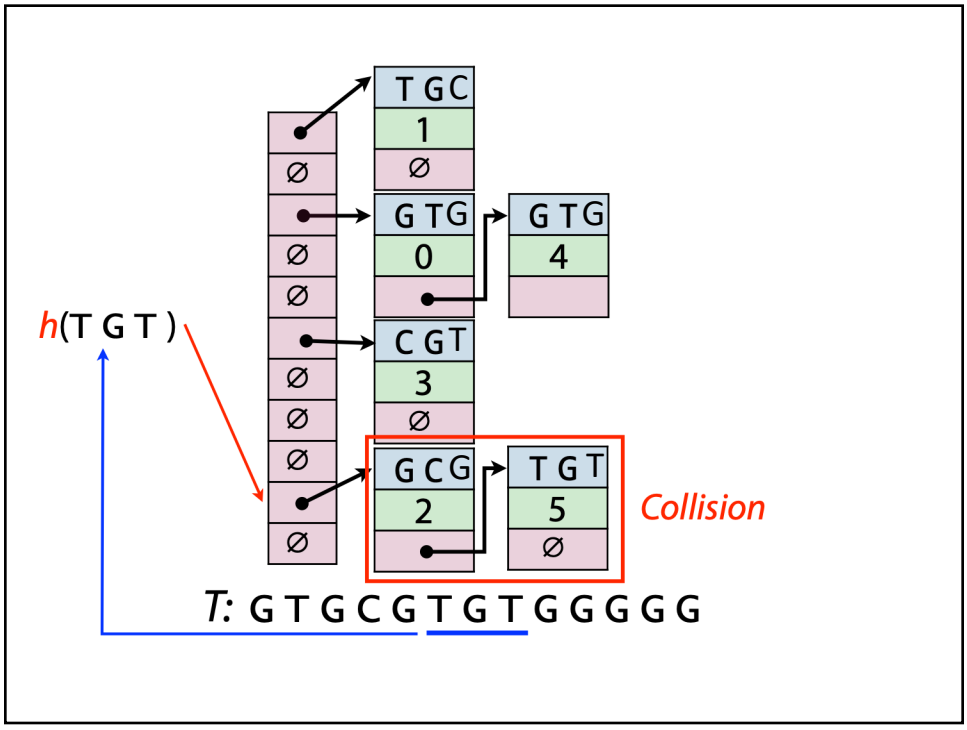
213


T G C
1
∅
G T G
0
∅
C G T
3
∅
G C G
2
∅
h(C G T)
T: G T G C G T G T G G G G G

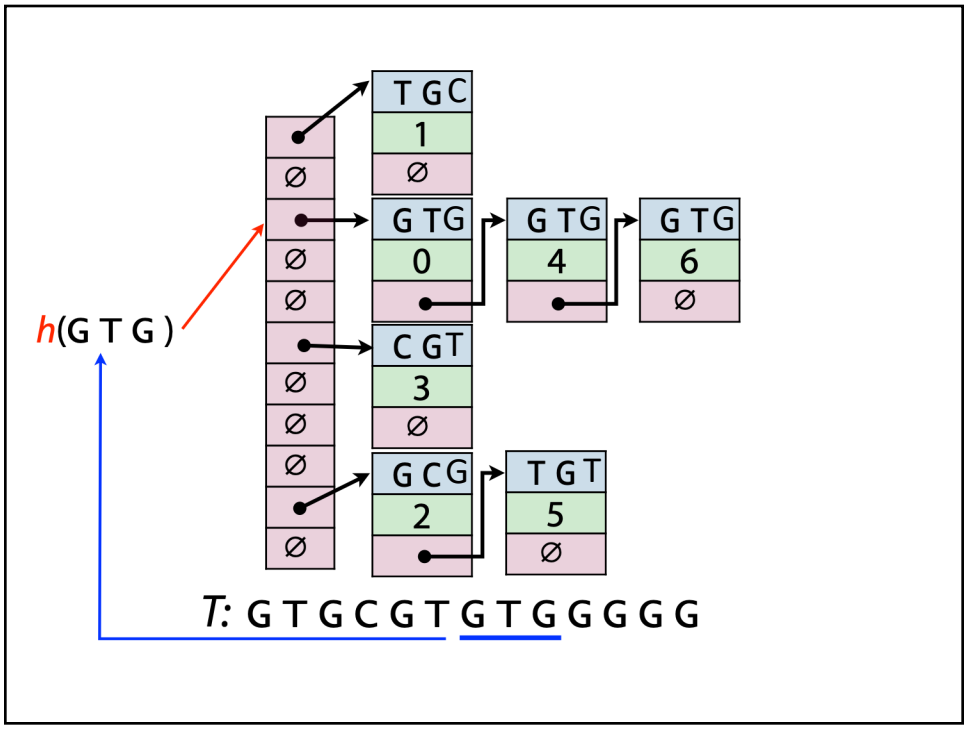
214


T G C
1
∅
G T G
0
G T G
4
∅
C G T
3
∅
G C G
2
∅
h(G T G)
T: G T G C G T G T G G G G

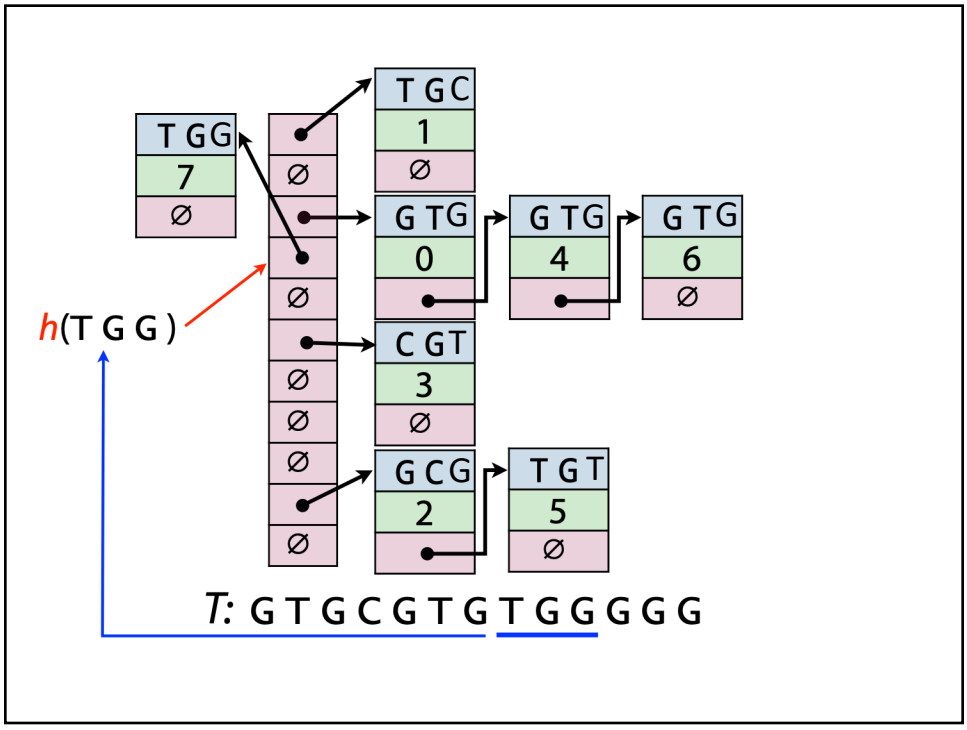
215


T G C
1
∅
G T G
0
G T G
4
C G T
3
∅
G C G
2
T G T
5
∅
Collision
h(T G T)
T: G T G C G T G T G G G G

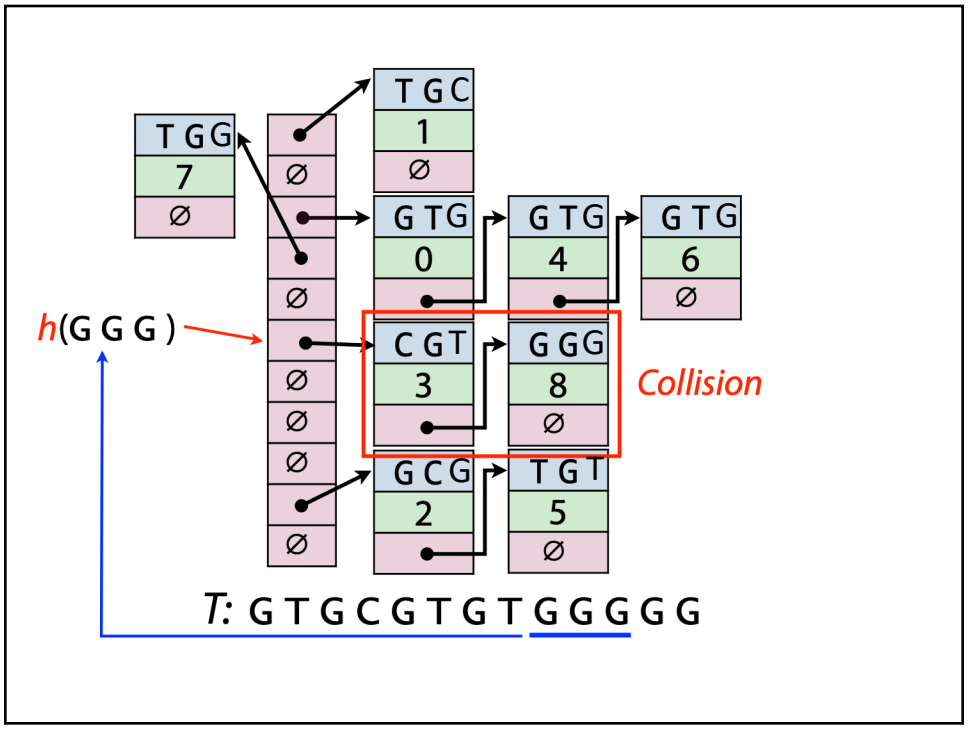
216


T G C
1
∅
G TG
0
G TG
4
G TG
6
∅
C GT
3
∅
G CG
2
T G T
5
∅
h(G T G)
T: G T G C G T G T G G G G

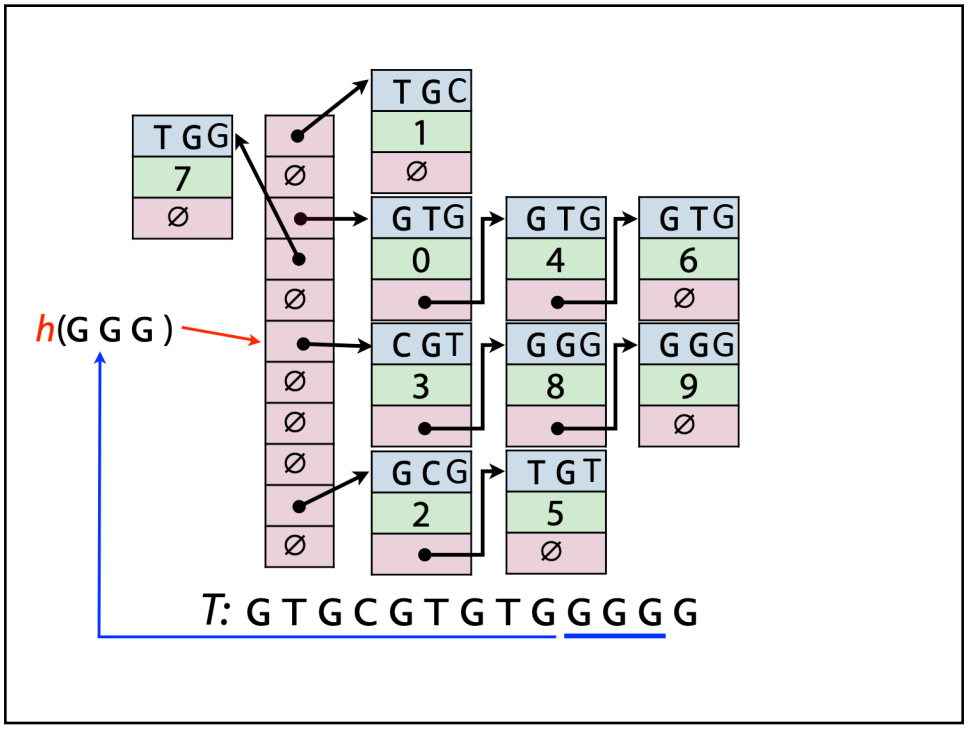
217


T GG
7
∅
T G C
1
∅
G TG
0
G TG
4
G TG
6
∅
C GT
3
∅
G CG
2
T G T
5
∅
h(T G G)
T: G T G C G T G T G G G G

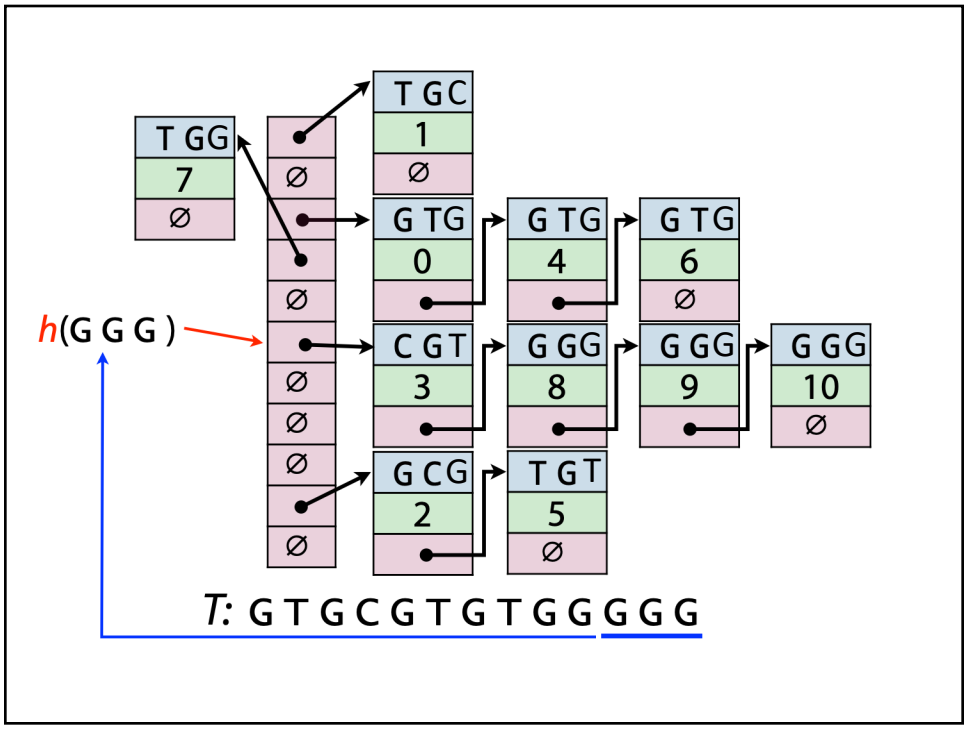
218


T G C
1
∅
T G G
7
∅
G T G
0
G T G
4
G T G
6
∅
h(G G G)
C G T
3
G G G
8
∅
Collision
G C G
2
T G T
5
∅
T: G T G C G T G T G G G G G

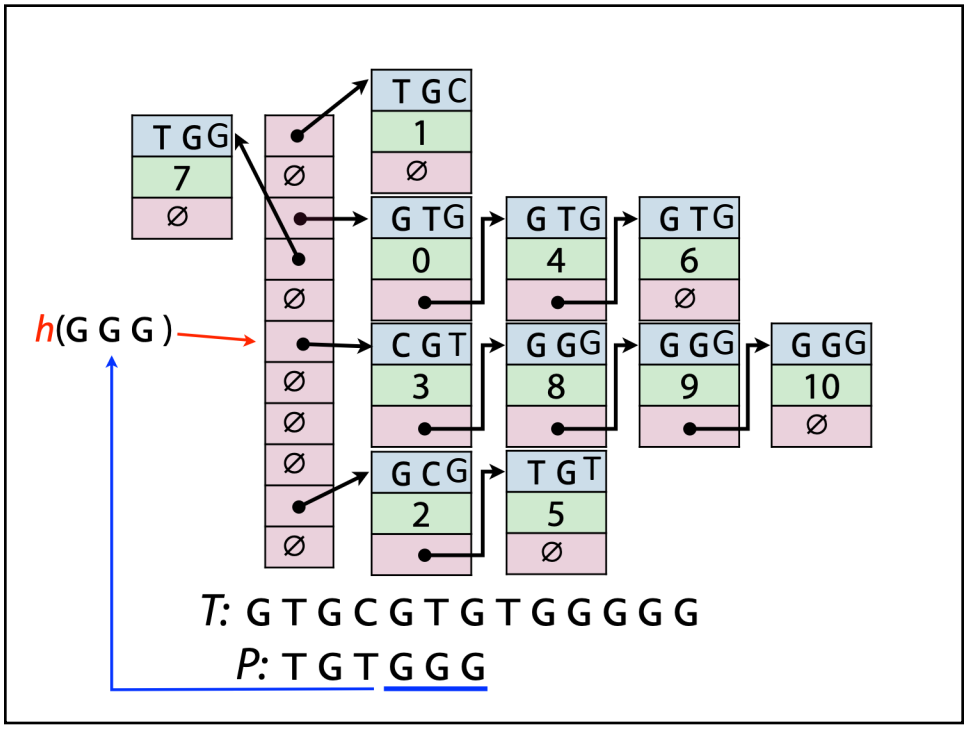
219


T G C
1
∅
T G G
7
∅
G T G
0
G T G
4
G T G
6
∅
h(G G G)
C G T
3
G G G
8
G G G
9
∅
G C G
2
T G T
5
∅
T: G T G C G T G T G G G G G

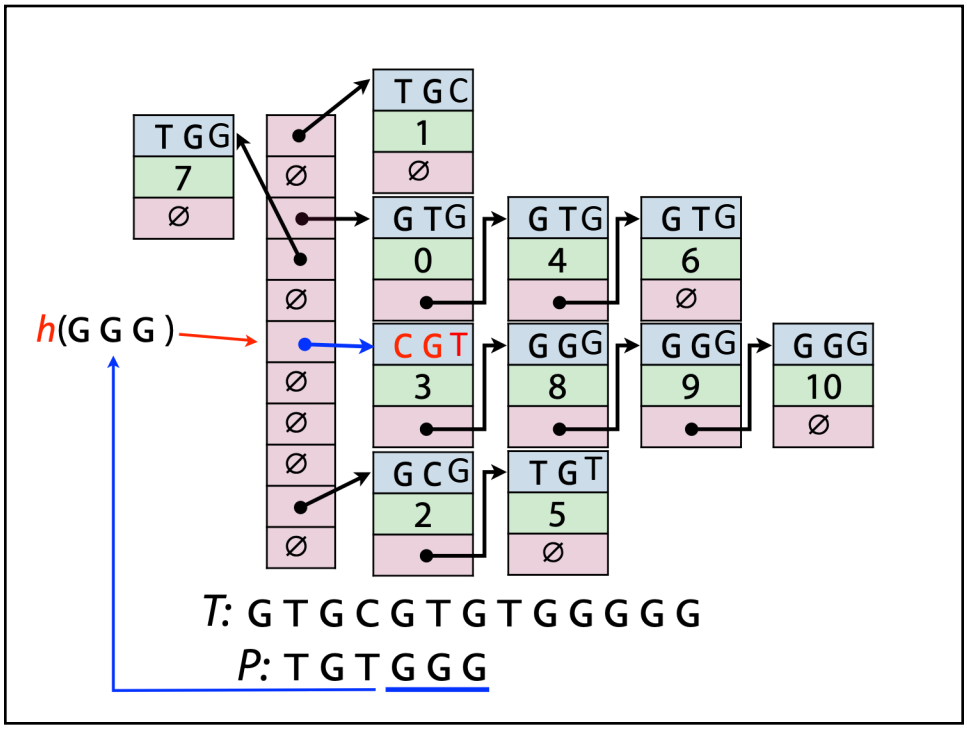
220


T GG
7
∅
T GC
1
∅
G TG
0
G TG
4
G TG
6
∅
C GT
3
G GG
8
G GG
9
G GG
10
∅
G CG
2
T GT
5
∅
h(G G G)
T: G T G C G T G T G G G G

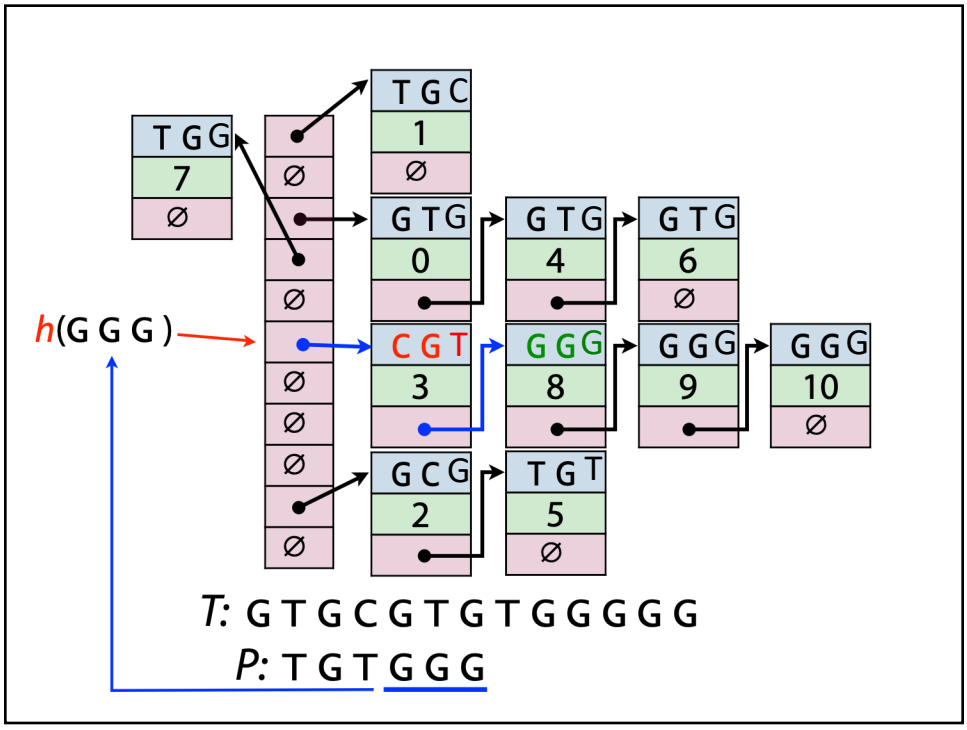
221


T GG
7
∅
T GC
1
∅
G TG
0
G TG
4
G TG
6
∅
C GT
3
G GG
8
G GG
9
G GG
10
∅
G CG
2
T GT
5
∅
h(G G G)
T: G T G C G T G T G G G G G
P: T G T G G G

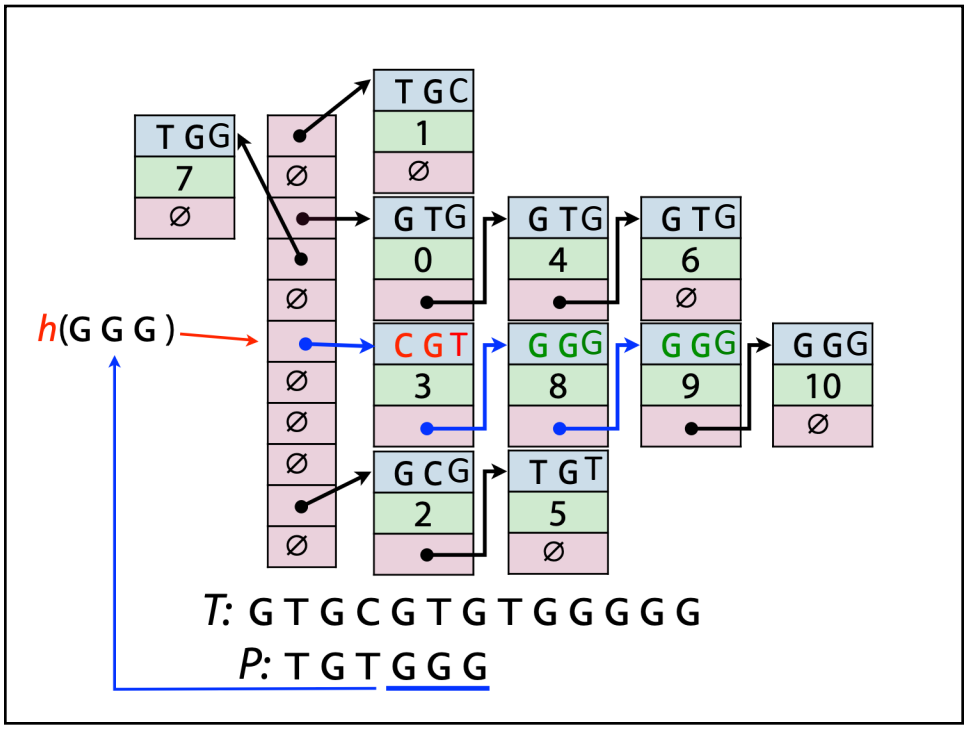
222


T G C
1
∅
T G G
7
∅
G T G
0
G T G
4
G T G
6
∅
h(G G G)
C G T
3
G GG
8
G GG
9
G GG
10
∅
G C G
2
T G T
5
∅
T: G T G C G T G T G G G G G
P: T G T G G G

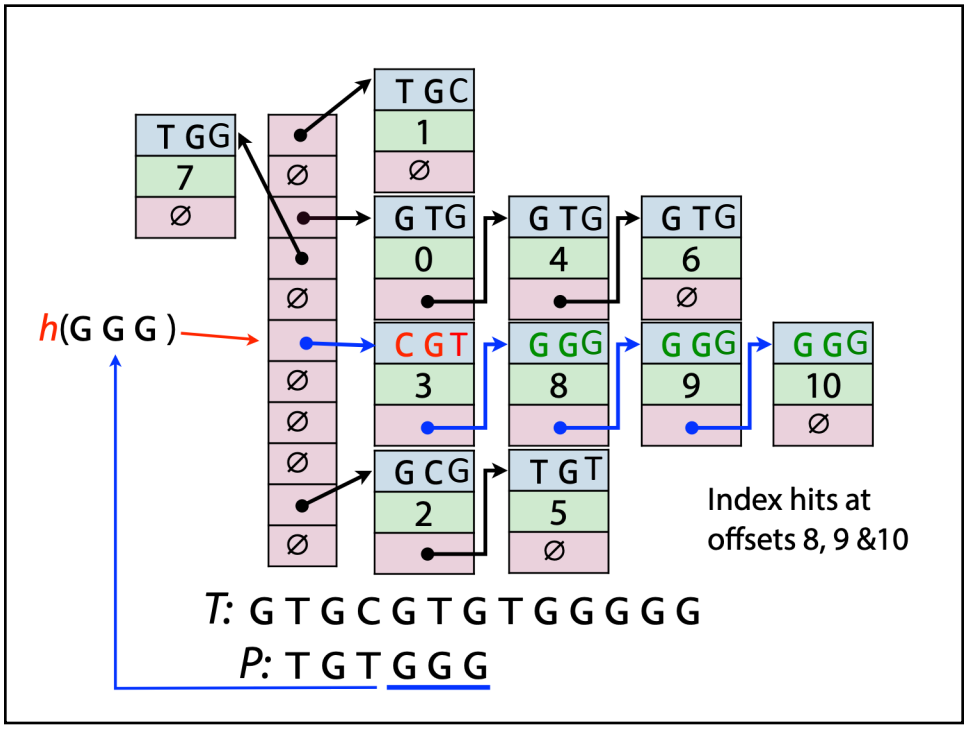
223


T G C
1
∅
T G G
7
∅
G T G
0
G T G
4
G T G
6
∅
h(G G G)
C G T
3
G GG
8
G GG
9
G GG
10
∅
G C G
2
T G T
5
∅
T: G T G C G T G T G G G G G
P: T G T G G G

224


T GG
7
∅
T GC
1
∅
G TG
0
C GT
3
G CG
2
G TG
4
G GG
8
T GT
5
∅
G TG
6
∅
G GG
9
G GG
10
∅
h(G G G)
T: G T G C G T G T G G G G G
P: T G T G G G

225


T GG
7
∅
T GC
1
∅
G TG
0
C GT
3
G CG
2
G TG
4
G GG
8
T GT
5
∅
G TG
6
∅
G GG
9
G GG
10
∅
h(G G G)
Index hits at
offsets 8, 9 &10
T: G T G C G T G T G G G G G
P: T G T G G G

226